AARON R. WILLIAMS

Email: aaronrwilliams100@gmail.com — Web: aaronrwilliams.com — Updated: August 6, 2026

Summary

Lead Data Scientist for Statistical Computing with expertise in machine learning, AI, privacy-enhancing technologies, administrative data, official statistics, and data governance. Principal and co-principal investigator on more than $3 million in funded research. Researcher and educator focused on developing next-generation technologies with Python and R, and training future professionals to advance evidence-based decision-making.

Education

Georgetown University - Washington, DC *May 2021*
M.S. Mathematics and Statistics

Virginia Commonwealth University - Richmond, VA *Dec. 2014*
Honors B.S. Economics

Virginia Commonwealth University - Richmond, VA *Dec. 2014*
Honors B.A. Music

Professional Experience

Urban Institute *June 2016 - Present*
Lead Data Scientist for Statistical Computing *Jan. 2024 - Present*
Senior Data Scientist *Jan. 2021 - Dec. 2023*
Data Scientist *Jan. 2019 - Dec. 2020*
Research Analyst *Jan. 2018 - Dec. 2018*
Research Assistant *Jun. 2016 - Dec. 2017*

Georgetown University McCourt School of Public Policy *Jan. 2020 - Present*
Adjunct Professor *Aug. 2021 - Present*
Instructor of Record *Jan. 2020 - Dec. 2020*

The Commonwealth Institute for Fiscal Analysis - Richmond, VA *June 2015 - May 2016*
Research Assistant

Grants

National Center for Science and Engineering Statistics *Sept. 2025 - Aug. 2026*
National Science Foundation — Subcontract — $200,000
Project: Implementing the Secure Data Statistics Platform for secure multiparty computing in the National Secure Data Service Demonstration project
Role: Principal Investigator

National Center for Science and Engineering Statistics *Sept. 2025 - Aug. 2026*
National Science Foundation — Subcontract — $250,000
Project: Providing Customer Service for a National Secure Data Service Through a Data Concierge
Role: Co-Principal Investigator

Bill and Melinda Gates Foundation *Nov. 2024 - June 2026*
Bill and Melinda Gates Foundation — Investment ID INV-071365 — $375,000
Project: Expand SLDS Data Access with Synthetic Data Generation & Validation
Role: Co-Principal Investigator

Georgetown University *Oct. 2024 - Mar. 2027*
Massive Data Institute — $100,000
Project: Validation Server Pilot for State Longitudinal Data Systems
Role: Co-Principal Investigator

National Center for Science and Engineering Statistics *Oct. 2023 - Nov. 2025*
National Science Foundation — $800,000
Project: Advancing Communication, Collaboration, and Evidence-building through Synthetic Survey of Earned Doctorates (ACCESS)
Role: Co-Principal Investigator

Department of Labor *Sept. 2022 - Feb. 2025*
Chief Evaluation Office — $1,524,088
Project: Disclosure Protection for the Department of Labor's Restricted-Use Data Access Program
Role: Co-Principal Investigator

Honors & Awards

2024 VCU Alumni's 10 Under 10 Award *Oct. 2024*
Virginia Commonwealth University

2023 President's Award for Evidence *Apr. 2024*
Urban Institute

2023 President's Award for Technology Innovation in Research & Data Science *Apr. 2024*
Urban Institute

2022 President's Award for Technology Innovation in Research & Data Science *Apr. 2023*
Urban Institute

2021 President's Award for Technology Innovation in Research & Data Science *Apr. 2022*
Urban Institute

2020 President's Award for Entrepreneurship *Apr. 2021*
Urban Institute

2020 Joint Statistical Meetings Student Travel Award *Mar. 2020*
Washington Statistical Society of the American Statistical Association

2017 President's Award for Technology Innovation in Research & Data Science *Apr. 2018*
Urban Institute

Student of the Year *May 2015*
Virginia Commonwealth University Economics Department

Dossier Award *Dec. 2014*
Virginia Commonwealth University Honors College

Provost Scholarship (All Tuition and Fees) *Aug. 2010 - May 2014*
Virginia Commonwealth University

Publications

Book Chapters

2. Bowen, C., Snoke, J., **Williams, A. R.**, & Barrientos, A. F. (2026). The Case for Researching Applied Privacy Enhancing Technologies. In *Data Privacy Protection and the Conduct of Applied Research*. University of Chicago Press

1. Bowen, C., Burman, L. E., McClelland, R., & **Williams, A. R.** (2024). Safe Data Technologies: Safely Expanding Access to Administrative Tax Data. In *Handbook of Sharing Confidential Data* (pp. 294–312). Chapman; Hall/CRC

Refereed Publications

10. Panavas, L., Snoke, J., Tyagi, E., Bowen, C., & **Williams, A. R.** (2025). But Can You Use It? Design Recommendations for Practical Alternatives to Differentially Private Interactive Systems. *Harvard Data Science Review*, *7*(4). https://doi.org/10.1162/99608f92.a1c2b7cb

9. **Williams, A. R.**, Snoke, J., Bowen, CMK., and Barrientos, A. F. (2025). "Disclosing Economists' Privacy Perspectives: A Survey of American Economic Association Members on Differential Privacy and Data Fitness for Use Standards." *Harvard Data Science Review.* https://doi.org/10.1162/99608f92.a8fb0371.

8. Snoke, J., Bowen, CMK., **Williams, A. R.**, and Barrientos, A. F. (2023) "Incompatibilities Between Current Practices in Statistical Data Analysis and Differential Privacy." *Journal of Privacy and Confidentiality.*, 14(3). doi:10.29012/jpc.872.

7. Barrientos, A. F., **Williams, A. R.** , Snoke, J., and Bowen, CMK. (2023) "A Feasibility Study of Differentially Private Summary Statistics and Regression Analyses with Evaluations on Administrative and Survey Data" *Journal of the American Statistical Association*, 1-24. https://doi.org/10.1080/01621459.2023.2270795

6. **Williams, A. R.**, and Bowen, CMK. (2023) "The Promise and Limitations of Formal Privacy." *WIREs Computational Statistics*. e1615. doi.org/10.1002/wics.1615

5. Bowen, CMK., Bryant, V., Burman, L., Khitatrakun, S., McClelland, R., Mucciolo, L., Pickens, M., and **Williams, A. R.** (2022) "Synthetic Individual Income Tax Data: Promises and Challenges." *National Tax Journal*, 75(4), 767-790.

4. Bowen, CMK., Bryant, V., Burman, L., Czajka, J., Khitatrakun, S., MacDonald, G., McClelland, R., Mucciolo, L., Pickens, M., Ueyama, K., **Williams, A. R.**, Wissoker, D., and Zwiefel, N. (2022) "Synthetic Individual Income Tax Data: Methodology, Utility, and Privacy Implications." *Privacy in Statistical Databases: International Conference, PSD 2022, Paris, France, September 21–23, 2022*, Proceedings (pp. 191-204). Cham: Springer International Publishing.

3. Williams, L. J., **Williams, A. R.**, and O'Boyle, E. (2023) "Assessment of Path Model Fit: Evidence of Effectiveness and Recommendations for Use of the RMSEA-P." *Organizational Research Methods*.

2. **Williams, A. R.** (2021) "Data Visualization Case Studies for High-Dimensional Data Validation." *Stat.* Volume10, Issue1 December 2021. https://doi.org/10.1002/sta4.334

1. Bowen, CMK., Bryant, V., Burman, L., Khitatrakun, S., McClelland, R., Stallworth, P., Ueyama, K.,

Williams, A. R. (2020) "A Synthetic Supplemental Public Use File of Low-Income Information Return Data: Methodology, Utility, and Privacy Implications." *International Conference on Privacy in Statistical Databases*, pages 257-270. Springer, Cham.

Technical Reports and Working Papers

15. Bowen, C., Hu, J., Kinney, S., Seeman, J., Snoke, J., & **Williams, A. R.** (2026). Expanding Data Access through Synthetic Datasets and Verification Metrics: The SEDSyn-23 Pilot for the Survey of Earned Doctorates, Final Report (ADC-SEDSyn-23-N04). *America's Datahub Consortium*
14. Seeman, J., **Williams, A. R.**, & Bowen, C. (2025). Synthetic Data for the Nebraska Statewide Workforce & Educational Reporting System. *Urban Institute*
13. Tyagi, E., and Taylor, S., and MacDonald, G., and Tamaroff, D., and Miller, J., and Bowen, CMK., and **Williams, A. R.** (2024) "A Privacy-Preserving Validation Server Version 2.0." *Urban Institute*.
12. **Williams, A. R.**, and Barrientos, A. F. and Snoke, J. and Bowen, C (2024) "Benchmarking DP Linear Regression Methods for Statistical Inference." *Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches and their Consequences, Spring 2024*.
11. **Williams, A. R.**, and Andre, J. (2023) "Opt-In Disclosure Protection." *Urban Institute*.
10. **Williams, A. R.**, Zhong, M., and Braga, B. (2023) "Using Machine Learning to Estimate Racially Disaggregated Wealth Data at the Local Level." *Urban Institute*.
9. Solari, C., Robin, L., Chelidze, T., Fudge, K., **Williams, A. R.**, and Acs, G. (2022) "Boosting Upward Mobility: Metrics to Inform Local Action" *Urban Institute*.
8. Bowen, CMK., **Williams, A. R.**, and Pickens, M. (2022) "Decennial Disclosure: An Explainer on Formal Privacy and the TopDown Algorithm." *Urban Institute*.
7. Barrientos, A. F., **Williams, A. R.**, Snoke, J., and Bowen, CMK. (2022) "Differentially Private Methods for Validation Servers" *Urban Institute*.
6. Smith, K. E., **Williams, A. R.**, and Mudrazija, S. (2021) "Modeling Income in the Near Term 8: Final Report" *Urban Institute*.
5. Cosic, D., **Williams, A. R.**, and Steuerle, E. C. (2020) "Do People Work Longer When They Live Longer?" *Working Paper, Retirement and Disability Research Consortium*.
4. Bowen, CMK., Burman, L., Khitatrakun, S., MacDonald, G., McClelland, R., Stallworth, P., Ueyama, K., **Williams, A. R.**, Zwiefel, N. (2020) "A Synthetic Supplemental Public-Use File of Low-Income Information Return Data: Methodology, Utility, and Privacy Implications." *Urban Institute*
3. Cosic, D. and **Williams, A. R.**. (2019) "Lifetime Income & Costs of the LI50+" *Research Report for the AARP Foundation*.
2. Smith, K. E. and **Williams, A. R.**. (2019) "Modeling Income in the Near Term 8: Dataset Documentation" *Urban Institute*.
1. Smalligan, J., **Williams, A. R.**, and Boyens, C. (2019) "Social Security's Earliest Eligibility Age" *Urban Institute*.

Software Packages

4. tidysynthesis. R package for synthetic data generation, evaluation, and visualization.

3. syntheval. R package for evaluating synthetic data utility and quality.

2. urbnmapr. R package providing geographic boundary files and mapping utilities.

1. urbnthemes. R package providing Urban Institute visualization themes.

Data Products and Interactive Applications

5. 2026. Drive-Time to Social Security Field Offices. Interactive mapping tool measuring geographic access to Social Security Administration field offices and estimating the effects of office closures.
4. 2026. Upward Mobility Data Dashboard. Public-facing dashboard tracking conditions associated with upward mobility and racial equity across counties and cities in the United States.
3. 2025. Mapping Diaper Insecurity in the US. Interactive data tool visualizing county-level diaper insecurity and access to basic-needs services.
2. 2025. Financial Health and Wealth Dashboard. Interactive dashboard presenting local-level indicators of wealth, savings, debt, and financial well-being.
1. 2020. Fixing Social Security. Interactive policy simulation exploring the effects of alternative Social Security reform proposals on beneficiaries and program finances.

Invited Presentations

5. July 12, 2023. *Using Machine Learning to Estimate Wealth at the Local Level*. US Census Bureau ML/AI Affinity Group, Washington, DC.
4. April 13, 2023. *tidysynthesis*. UCLA Synthetic Data Conference, Los Angeles, CA.
3. August 9, 2022. *Working with Stakeholders to Protect Data Confidentiality*. Joint Statistical Meetings, Washington, DC.
2. January 28, 2022. *Fully synthetic microdata for public policy analysis*. International Privacy Day: New Developments in Synthetic Data Generation. American Statistical Association Privacy and Confidentiality Committee.
1. April 22, 2021. *Data Visualization for High-Dimensional Data Validation*. 14th annual Innovations in Design, Analysis and Dissemination (IDAD): Frontiers in Biostatistical and Data Science conference, Kansas City, MO.

Contributed Presentations

19. February 10, 2026. *Benchmarking Differential Privacy for Applied Public Policy Analysis*. Privacy and Public Policy Conference. Washington, DC.
18. August 3, 2024. *Reproducible Research Bootcamp*. 2024 Joint Statistical Meetings. Portland, OR.
17. June 17, 2024. *Navigating the Privacy-Utility Tradeoff: An Introduction to Data Privacy Techniques*. International Conference on Establishment Statsitics VII. Glasgow, Scotland.
16. May 17, 2024. *Benchmarking DP Linear Regression Methods for Statistical Inference*. NBER Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches, and the Consequences. Washington, DC.
15. November 14, 2023. *Using Machine Learning to Estimate Wealth at the Local Level*. US Census Bureau's Workshop on Advancing Research on Race, Ethnicity and Inequality, Washington, DC.

14. August 7, 2023. *Professional Development Course: Introduction to Data Privacy and Data Synthesis Techniques*. 2023 Joint Statistical Meetings. Toronto, ON.
13. May 4, 2023. *Disclosing Economists' Privacy Perspectives: A Survey of American Economic Association Members on Differential Privacy and Data Fitness for Use Standards*. NBER Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches, and the Consequences. Cambridge, MA.
12. April 20, 2023. *Boosting Upward Mobility from Poverty*. VCU Berglund Seminar, Richmond, Virginia.
11. February 8, 2023. *Using Machine Learning to Understand Wealth at the Local Level*. Census Virtual SIPP Conference.
10. July 28, 2022. *Introducing library(tidysynthesis)*. rstudio::conf(2022), National Harbor, MD.
9. June 3, 2021. *Synthetic Data Generation with tidysynthesis*. American Statistical Association's Symposium on Data Science and Statistics. St. Louis, MO.
8. November 18, 2020. *Future Plans for Secure Research Access to Administrative Tax Data*. National Tax Association annual meeting. Denver, Colorado. With Graham MacDonald.
7. August 5, 2020. *A Synthetic Supplemental Public Use File of Low-Income Information Return Data: Methodology, Utility, and Privacy Implications*. American Statistical Association 2020 Joint Statistical Meeting, Philadelphia, PA.
6. July 27, 2020. *Lessons Learned from Synthesizing IRS Tax Data*. Association for Public Data Users (APDU) Annual Conference, Arlington, VA. With Kyle Ueyama.
5. July 22, 2020. *Lifetime Income & Costs of the LI50+*. Bureau of Labor Statistics Consumer Expenditure Surveys (CE) Survey Methods Symposium and Microdata Users' Workshop, Washington, DC.
4. June 3, 2020. *Data Visualization for High-Dimensional Data Validation: Case Studies on Evaluating Microsimulation Modeling and Synthetic Data Generation*. American Statistical Association's Symposium on Data Science and Statistics, Pittsburgh, PA.
3. May 20, 2020. *Iterating Branded Fact Sheets and Web Pages with R Markdown*. Good Tech Fest, Chicago, IL.
2. March 28, 2020. *Iterating Branded Fact Sheets and Web Pages with R Markdown*. SatRday DC, Washington, DC.
1. February 7, 2020. *Generation X, Millennials, and the Future of Retirement*. VCU Berglund Seminar, Richmond, VA.

Professional Activities and Affiliations

Service:

Bureau of Labor Statistics

- Technical Advisory Committee 2024–2025

Journal Referee

- Harvard Data Science Review
- Journal of Survey Statistics and Methodology
- International Journal of Population Data Science

Symposium on Data Science and Statistics

- Co-organizer of the Data Visualization Track 2022

National Professional Memberships:

American Statistical Association

- Traveling Course Instructor 2022, 2023, 2024
  - Data Visualization in R
  - Offered full-day workshops to 15 chapters: Nebraska, Louisville, Columbus, Alaska, Philadelphia, Central Arkansas, North Carolina, Western Michigan, Nevada, New York, Princeton, Oklahoma, Houston, Cleveland, and North Texas.
- Member Mar. 2020 – Present

Blogs and Commentary

26. Walsh, J., **Williams, A. R.**, Fedorowicz, M., Kovalski, M. A., & Holland, N. (2026). Three Major Challenges to Boosting Upward Mobility from Poverty. *Urban Wire*
25. **Williams, A. R.** & Axelrod, J. (2026). Building an Agentic AI Tool to Support Our Upward Mobility Initiative Dashboard. *Data@Urban*
24. Axelrod, J., **Williams, A. R.**, Purohit, R., & Tyagi, E. (2026). What It Takes to Make Research and Policy Knowledge AI Ready. *Data@Urban*
23. Schwabish, J., **Williams, A. R.**, & Morrison, G. (2026). How We Used Routing Analysis to Map Access to Social Security Offices. *Data@Urban*
22. Bowen, C., & **Williams, A. R.** (2026b). Part 2: Is Data Privacy Dead? The Answer Must be No. *Association of Public Data Users*
21. Bowen, C., & **Williams, A. R.** (2026a). Part 1: Is Data Privacy Dead? The Answer Must be No. *Association of Public Data Users*
20. Tyagi, E., Reeves, M., **Williams, A. R.**, Zeehandelaar Shaw, D., & Bowen, C. (2026). Five Lessons for Building Sustainable Data Systems to Support Policy Insights . *Data@Urban*
19. Walsh, J., & **Williams, A. R.** (2026). Opportunities for Rewarding Work Vary Widely Within and Across Counties. *Urban Wire*
18. Bowen, C., & **Williams, A. R.** (2025c). Two Ideas to Support and Reward Data Collectors. *Association of Public Data Users*
17. Bowen, C., & **Williams, A. R.** (2025b). A Day in the Life with Federal Government Data. *Association of Public Data Users*
16. **Williams, A. R.** & Bowen, C. (2025). Driving While Looking Through a Shattered Rearview Mirror. *Association of Public Data Users*
15. Bowen, C., & **Williams, A. R.** (2025a). Eggs Don't Grow on Trees and Neither Does Data: How to Teach K-12 Students Where Data Comes From. *Association of Public Data Users*
14. Bowen, C., & **Williams, A. R.** (2025d). The Forgotten Egg: The Importance of Public Federal Data in the United States. *Data Rescue Project*
13. **Williams, A. R.**. (2023) "Improving Official Statistics with Opt-In Disclosure Protection." *Urban Wire*.
12. Zhong, M. and **Williams, A. R.**. (2022) "In Chicago, Neighborhoods Have Stark Differences in Economic Opportunity." *Urban Wire*.

11. Schwabish, J. and **Williams, A. R.**. (2021) "Creating Better, More Informative Map Legends in R." *Urban Wire*.
10. Schwabish, J., **Williams, A. R.** (2021) "Why Disability Insurance Participation Might Be Higher in Some Parts of the Country" *Urban Wire*.
9. Bowen, C., **Williams, A. R.**, & Narayanan, A. (2021). To Advance Racial Equity, Releasing Disaggregated Data while Protecting Privacy Will Be Key. *Urban Wire*
8. **Williams, A. R.** (2019) "Expanding the Earned Income Tax Credit Can Support Older Working Americans." *Urban Wire*.
7. Smalligan, J., **Williams, A. R.**, and Boyens, C. (2019) "If the Early Retirement Age Changes, At-Risk Older Workers Will Need More Support." *Urban Wire*.
6. Chartoff, B., Kelly, J., MacDonald, G., Schwabish, J., **Williams, A. R.**. (2019) "Scripting versus Point-and-Click, a Slack Debate" *Data@Urban*.
5. Schwabish, Jon and **Williams, A. R.**. (2019) "Learning R: A Conversation." *Data@Urban*.
4. **Williams, A. R.** (2019) "Building an R Community at the Urban Institute." *Data@Urban*.
3. **Williams, A. R.** (2018) "Iterated PDFs with R Markdown." *Data@Urban*.
2. **Williams, A. R.** (2018) "Iterated fact sheets with R Markdown." *Data@Urban*.
1. **Williams, A. R.** (2018) "Why the Urban Institute Visualizes Data with ggplot2." *Data@Urban*.